

MID-II Assignment: Python Programming

Instructions

- Each student/group (maximum 3 members) must select **at least one project**.
- Students may undertake more than one project; **additional projects will carry additional marks** based on quality and completion.
- Proper documentation and project demonstration are mandatory.
- Code should be modular, well-structured, and properly commented.
- All code must be uploaded to GitHub.
- Viva will be conducted based on implementation and conceptual understanding.

BCA IV Semester

1. Student Management System

- Add, update, delete, and search student records
- Store data using CSV/JSON files
- Sort records by name or marks
- Admin login authentication
- Generate reports (topper, average marks, etc.)
- Implement exception handling for invalid inputs

2. Library Management System

- Book issue and return with due date tracking
- Fine calculation for late returns
- Search books by title or author
- Maintain user and book records
- Display available and issued books
- Backup and restore data

3. Quiz Application

- Random question selection from file/database
- Timer-based quiz system
- Negative marking scheme
- Leaderboard generation
- Optional GUI using Tkinter
- Category-based quiz support

4. Bank Account Simulation

- Support for multiple account types
- Transaction history with timestamps
- Interest calculation
- Secure PIN-based login
- Fund transfer functionality
- Export transaction history

5. File Organizer Tool

- Automatic file categorization
- Duplicate file detection
- Batch file renaming using patterns
- Optional GUI interface
- Maintain operation logs
- Undo last operation feature

1. Fake News Detection

- Text preprocessing (stopword removal, stemming, lemmatization)
- Feature extraction using TF-IDF
- Implementation of multiple ML models
- Hyperparameter tuning
- Performance evaluation using standard metrics
- Deployment using Flask or Streamlit

2. Customer Segmentation

- Data preprocessing and normalization
- Clustering using K-Means, DBSCAN, and Hierarchical methods
- Elbow method and Silhouette score analysis
- Visualization of clusters
- Derivation of business insights
- Interactive dashboard using Streamlit

3. Credit Card Fraud Detection

- Handling imbalanced data (e.g., SMOTE)
- Feature selection techniques
- Comparison of multiple models
- ROC-AUC and performance analysis
- Real-time fraud detection simulation
- Model explainability using SHAP/LIME

4. Stock Price Prediction

- Time-series data preprocessing
- Implementation of LSTM/GRU models
- Comparison with traditional models
- Hyperparameter tuning
- Visualization of predictions
- Web-based prediction interface

5. Recommender System

- Content-based and collaborative filtering
- Matrix factorization techniques (e.g., SVD)
- Evaluation using RMSE/MAE
- Handling cold-start problem
- User interface development
- Personalized recommendation dashboard